

Customer Shopping Behavior Analysis

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

Dataset Summary

The dataset contains comprehensive information across 3,900 transactions with 18 key columns capturing multiple dimensions of customer behavior and purchase patterns.

Dataset Size

3,900 rows

18 columns

Data Quality

37 missing values

In Review Rating column

Key Features include:

- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python to ensure data quality and consistency:

1 Data Loading

Imported the dataset using pandas.

2 Initial Exploration

Used `df.info()` to check structure and `.describe()` for summary statistics.

3 Missing Data Handling

Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

4 Column Standardization

Renamed columns to **snake case** for better readability and documentation.

5 Feature Engineering

Created **age_group** column by binning customer ages and **purchase_frequency_days** column from purchase data.

6 Data Consistency Check

Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.

7 Database Integration

Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

Revenue Analysis by Customer Segments

We performed structured analysis in PostgreSQL to answer key business questions about revenue generation across different customer demographics and behaviors.

Revenue by Gender

Compared total revenue generated by male vs. female customers:

Gender	Revenue (USD)
Female	\$75,191
Male	\$157,890

Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status:

Status	Customers	Avg Spend
Yes	1,053	\$59.49
No	2,847	\$59.87

Product Performance and Customer Preferences

Analysis revealed clear patterns in product ratings, purchasing behavior, and category preferences that inform strategic product positioning.

Top 5 Products by Rating

Found products with the highest average review ratings:

Item Purchased	Average Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Discount-Dependent Products

Identified 5 products with the highest percentage of discounted purchases:

Item Purchased	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation and Behavior Patterns

Classified customers into distinct segments based on purchase history and analyzed their subscription adoption rates.

3116

Loyal Customers

Repeat purchasers with established buying patterns

701

Returning Customers

Customers with multiple purchases but less frequency

83

New Customers

First-time or early-stage purchasers

Repeat Buyers & Subscriptions

Checked whether customers with >5 purchases are more likely to subscribe:

Subscription Status	Repeat Buyers
No	2,518
Yes	958

Category and Shipping Analysis

Examined top-performing products within each category and compared purchasing behavior across different shipping methods.

Top 3 Products per Category

Listed the most purchased products within each category:

Rank	Category	Item	Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

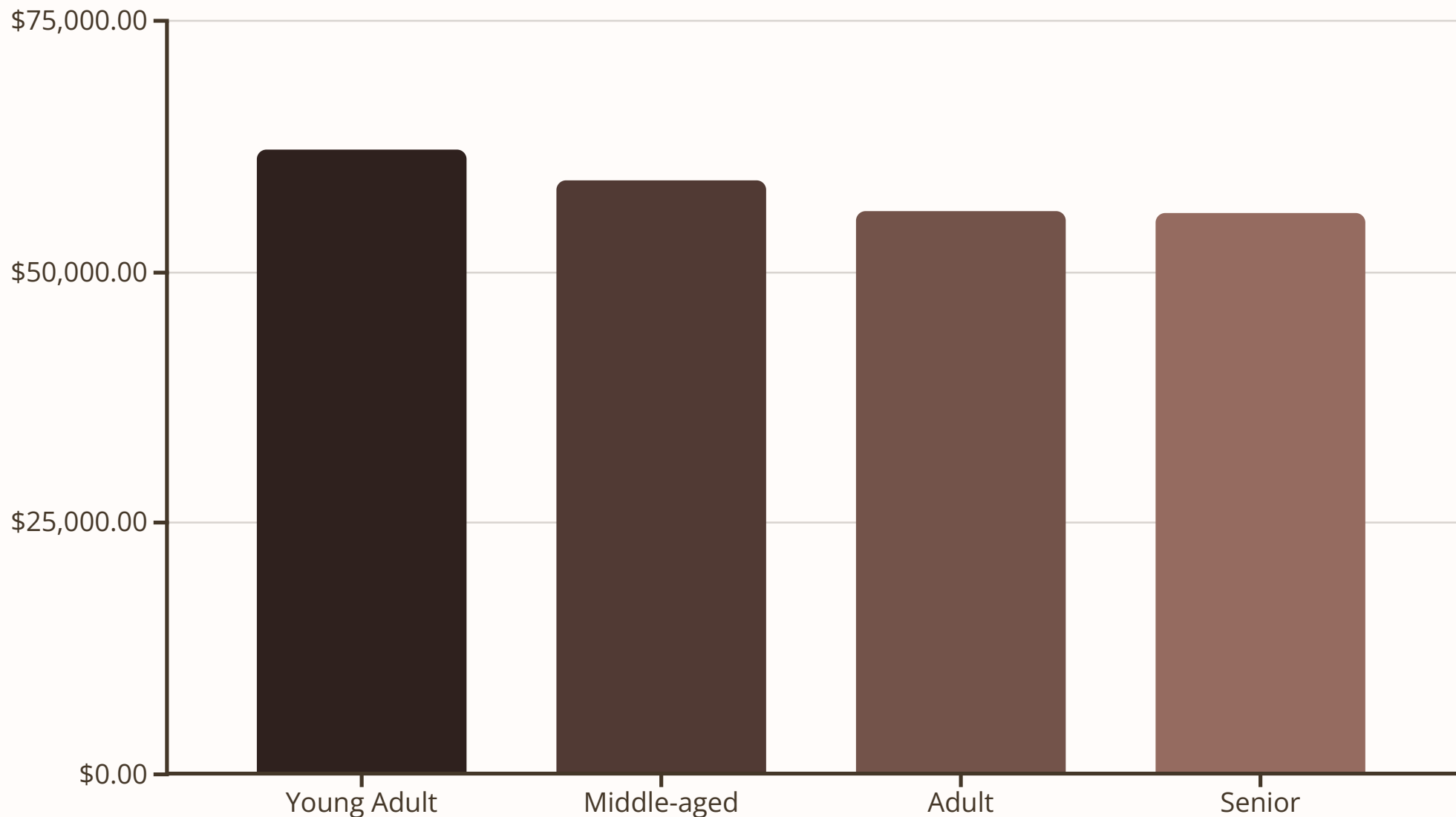
Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping:

Shipping Type	Avg Amount (USD)
Standard	\$58.46
Express	\$60.48

Revenue by Age Group

Calculated total revenue contribution of each age group to identify which demographic segments drive the most value for the business.



Young Adult customers generate the highest total revenue at \$62,143, followed by Middle-aged customers at \$59,197. Adult and Senior age groups contribute nearly equally at \$55,978 and \$55,763 respectively, indicating relatively balanced revenue distribution across mature customer segments.

Interactive Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually and enable real-time monitoring of key business metrics.



Business Recommendations

Based on comprehensive analysis of customer shopping behavior, we recommend the following strategic initiatives:

Boost Subscriptions

Promote exclusive benefits for subscribers to increase adoption rates and create recurring revenue streams.

Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment and increase customer lifetime value.

Review Discount Policy

Balance sales boosts with margin control by analyzing discount-dependent products and optimizing discount strategies.

Product Positioning

Highlight top-rated and best-selling products in campaigns to maximize visibility and conversion rates.

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users to optimize marketing spend and ROI.